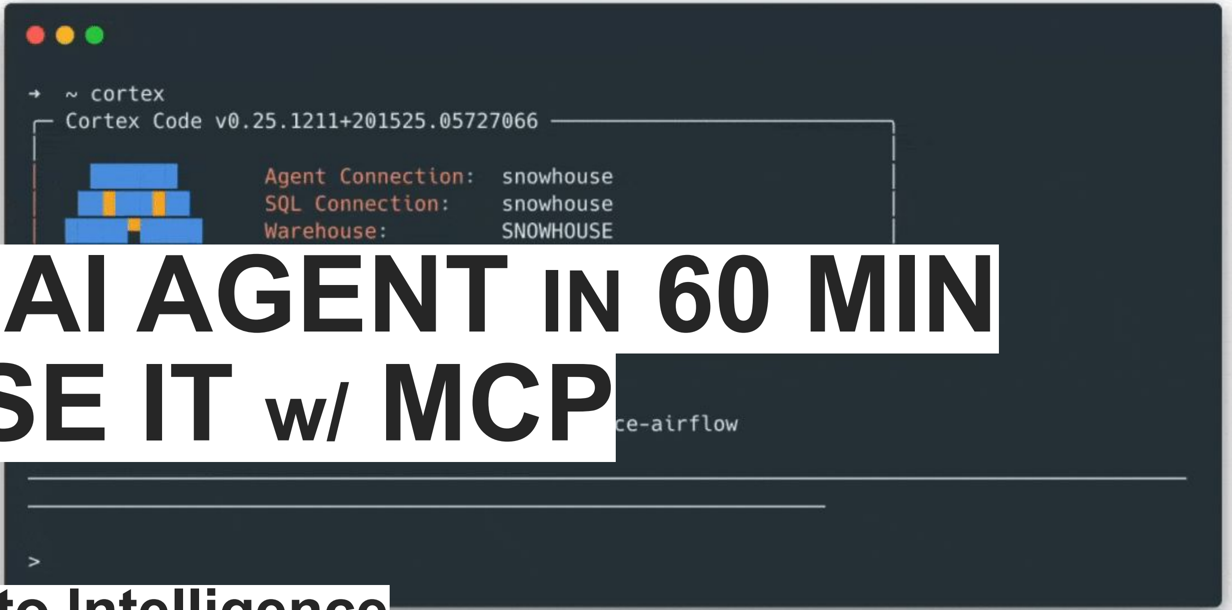
BUILD AN AI AGENT IN 60 MIN then EXPOSE IT w/ MCP



From Foundation to Intelligence

Build, Extend, & Expose a PROD AI Agent w/ CoCo

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Snowflake | July 28, 2026

Tonight

- 0–10 min | Why CoCo + The Stack
- 10–20 min | Step 0: AGENTS.md + CLI setup
- 20–42 min | Build GitTrend agent (5 prompts)
- 42–60 min | Step 3: Wire the MCP Server
- 60–68 min | Run It: query from Claude Desktop | Cursor
- 68–75 min | Q&A + take-home

What You'll Walk Away w/

By the end of tonight you will have built, deployed, and connected a PROD AI agent.

1. A Working AI Agent

- Real, queryable, yours to keep.
- 107M real GitHub events.
- Answers natural language questions.

2. An MCP Endpoint

- Any tool in the world can query it.
- Claude Desktop, Cursor, VS Code.
- Same agent. Every client.

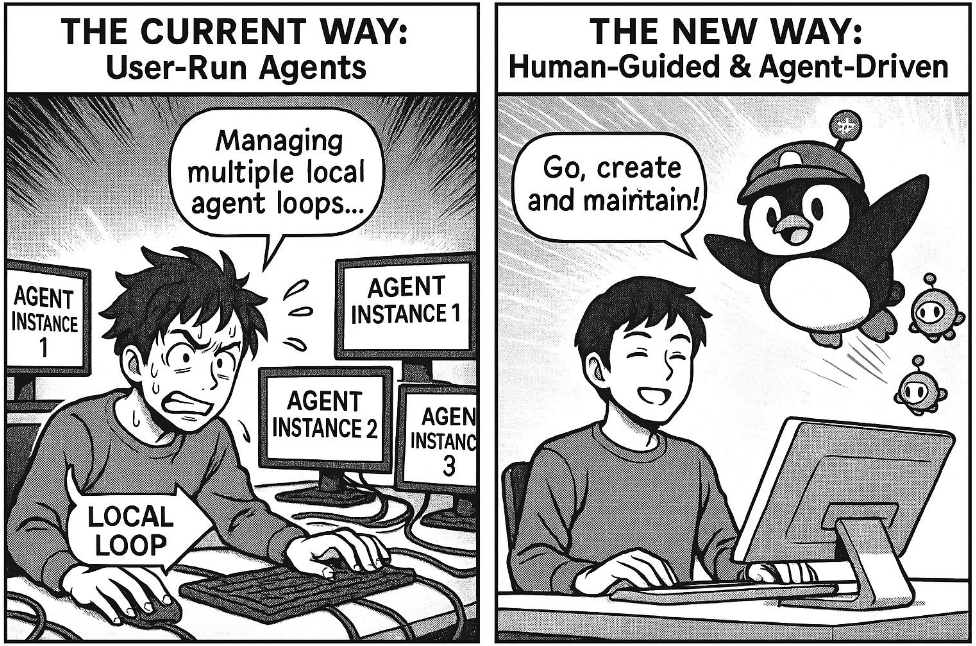
3. The Mental Model

- AGENTS.md → CoCo reads your context
- CoCo → writes every SQL statement
- Cortex AI → search + complete + agent
- MCP Server → exposes it to the world

- Same 5 steps work on any dataset.
- GitHub is how we learn it tonight.

THE PROBLEM

The world is changing: Truly AI everywhere



Development is moving from human-driven to human-*guided* & agent-driven.

General AI Tools Weren't Built for Data Teams

Your data is the moat. Tonight we build the agent that reaches it.

They're coding-only.

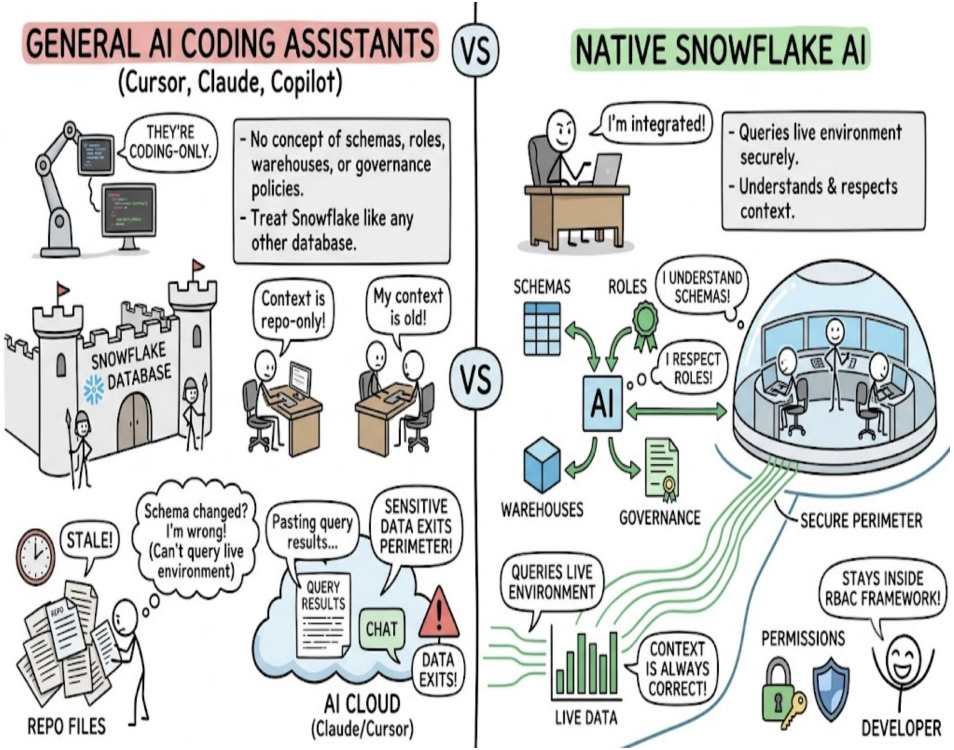
Cursor, Claude Code, and Copilot complete code.
They have no concept of schemas, roles, warehouses, or governance policies > treat Snowflake like any other database.

Their context is repo-only: & goes stale.

They read files you give them. The moment your schema changes, their context is wrong. They can't query your live environment.

Data leaves your secure perimeter.

Pasting query results into Claude Code or Cursor means sensitive data exits your governance boundary.
With CoCo, everything stays inside Snowflake's RBAC framework.



CoCo: The Data-Native Agent Harness

Same CLI-native pattern as Claude Code: but Snowflake-native. The difference is the harness.

Layer 1: Context (SnowScope)

- CoCo uses SnowScope: tuned to find the right table, semantic view, and tool before you ask.
- Claude Code has raw MCP access. CoCo has the trained instinct to use it correctly.

Layer 2: Skills

- Native dbt, Airflow, and Databricks skills built into the harness.
- Fire at the right moment. No configuration needed.

Layer 3: Governance

- Your data never leaves Snowflake.
- Same RBAC. No caching by model providers.
- Not vendor-locked: Claude, GPT, Grok, Gemini all supported.

Per-tool positioning

- Cursor users: CoCo runs inside Cursor as a terminal agent.
- Claude Code users: same CLI pattern — but Snowflake-native.
- Copilot users: CoCo is an agent that executes end-to-end, not just completes.

Proven, Not Promised

72.1% on ADE Bench vs 65.1% for Claude Code: pipeline fixes from days to hours, across 7,100+ accounts.

72.1%

- **ADE-Bench Pass Rate**
- #1 on ADE-Bench
- vs 65.1% Claude Code
- vs 65.1% Codex
- Same model — harness wins

51%

Fewer Tokens

- vs Claude Code
- 8% fewer steps
- Faster. Cheaper.
- Less context saturation

5x

More PRs Shipped

- In the wild
- Pipeline fixes: days → hours

$0

- **Extra Cost**
- Included w/ Snowflake
- vs $20–40/user/mo for Cursor or Claude Code

THE STACK

Your Agent Stack Tonight

Six layers. 75 minutes. Built entirely by CoCo. MCP Server is the new step for v2.

FOUNDATION

① AGENTS.md

- Persistent context file
- CoCo reads every session

② GITTREND_DB

- 107M real GitHub events
- Loaded from public S3

INTELLIGENCE

③ V_TRENDING_AI_REPOS

- View: stars by AI repo
- Last 30 days of data
-

④ AI_COMPLETE

- SQL results → language
- claude-sonnet-4-6

⑤ GITHUB_REPO_SEARCH

- Cortex Search Service

EXPOSURE

⑥ GITTREND Agent

- Model: auto
- search + data_to_chart tools

⑦ GITTREND_MCP

- MCP Server: NEW tonight
- Claude Desktop | Cursor

⑧ OAuth Integration

- Auth for MCP clients

LET'S BUILD

https://github.com/sfc-gh-rbachala/building-ai-agents-with-coco-workshop

http://bit.ly/441DbCC

Step 0: AGENTS.md + CLI Connection

⏱️ **Do this before everything else**

Create AGENTS.md in your project folder

account: <orgname>-<accountname>
role: ACCOUNTADMIN
warehouse: WORKSHOP_WH
database: GITTREND_DB
schema: GITTREND_DB.PUBLIC

Why AGENTS.md matters

CoCo reads this file at the start of every session.
It knows your account, warehouse, and constraints > without you re-explaining.
Under 200 lines keeps CoCo compliance near 100%.

Start CoCo

Run: cortex (from inside the gittrend-workshop/ folder)
First launch: setup wizard handles connection.

Once in, CoCo confirms AGENTS.md loaded in the session header.

Step 1 — Load the Data

⏱️ | **Fire this prompt. Move immediately.**

Paste this prompt into CoCo:

"Run the setup SQL: create GITTREND_DB, WORKSHOP_WH,
load 107M GitHub events from s3://sfquickstarts/vhol_building_ai_agents_with_coco/
into GITHUB_EVENTS table. Enable cross-region Cortex. Verify the count."

What CoCo does

Creates GITTREND_DB, WORKSHOP_WH (Small, auto-suspend 60s)
Enables CORTEX_ENABLED_CROSS_REGION = ANY_REGION
Creates a JSON stage + GITHUB_EVENTS table
Runs COPY INTO: ~4 min on Small warehouse

This is 30 days of real GitHub events >> ~107M rows

WatchEvents, PushEvents, ForkEvents, IssueEvents.
Checkpoint: SELECT COUNT(*) returns ~107,752,158
Stuck? → CHECKPOINTS.sql → SETUP block

Step 2 — Build the GitTrend Agent

⏱️: 5 prompts. 4 objects. 1 agent. | Stuck? → CHECKPOINTS.sql

① Explore schema

- Prompt: explore GITTREND_DB.PUBLIC.GITHUB_EVENTS, find the right columns for star activity
- Checkpoint: WatchEvent = star, REPO_NAME + CREATED_AT

② Build V_TRENDING_AI_REPOS view

- Prompt: trending AI repos, last 30 days, min 10 stars, top 20
- ★ Discovery moment: "What's at the top of your list?" Let it land.

③ Add AI_COMPLETE summary

- Prompt: wrap view results in AI_COMPLETE, 3–4 sentence summary
- Note: use AI_COMPLETE — CORTEX.COMPLETE is deprecated (EOL 2026)

④ Create Cortex Search Service

- Prompt: GITHUB_REPO_SEARCH on description, WORKSHOP_WH, 1hr lag
- Fire and immediately go to ⑤ — they overlap.

⑤ Create GITTREND Agent

- Prompt: model=auto, GITHUB_REPO_SEARCH as search tool,
- data_to_chart tool, system prompt = GitHub trend analyst

Bonus: ask GitTrend for a bar chart

- "Show me a bar chart of the top 10 repos by stars"
- data_to_chart fires automatically — no extra config needed.
- Checkpoint: SHOW AGENTS → GITTREND active

Step 3 — Wire the MCP Server

⏱️ | New material: slow down here.

Part 1: Create MCP Server object

- Prompt: CREATE MCP SERVER GITTREND_DB.PUBLIC.GITTREND_MCP
- exposes GITTREND as CORTEX_AGENT_RUN tool
- Verify: SHOW MCP SERVERS IN SCHEMA GITTREND_DB.PUBLIC

Part 2: Set up OAuth

Run SQL: CREATE SECURITY INTEGRATION GITTREND_MCP_OAUTH
Then: SELECT SYSTEM$SHOW_OAUTH_CLIENT_SECRETS(...)

Save OAUTH_CLIENT_ID and OAUTH_CLIENT_SECRET
Set your DEFAULT_ROLE + DEFAULT_WAREHOUSE (ALTER USER)

Stuck? → CHECKPOINTS.sql → Checkpoint 6

Part 3: Connect your MCP client

- MCP Server URL: https://<account-url>/api/v2/databases/GITTREND_DB/schemas/PUBLIC/mcp-servers/GITTREND_MCP
- ⚠️ Use hyphens (-) not underscores (_) in account URL
- claude.ai: Settings → Connectors → Add custom connector
- Cursor: edit ~/.cursor/mcp.json, add mcpServers block
- CoCo Desktop: auto-discovers — no OAuth setup needed

Fallback — no MCP client yet?

- Left nav → AI & ML → Agents → GITTREND → Preview in Snowflake CoWork

RUN IT

Ask Your Agent From Your AI Tool

⏱️ | **Slow way down. Let it land.**

Open claude.ai, Cursor, or CoCo Desktop. GitTrend should be in your tools list.

"What's the fastest-growing AI project in the last 30 days?"

That answer is grounded in 107 million real GitHub events.
You built the agent. You built the endpoint.

Ask follow-ups — GitTrend remembers the conversation (Agent Threads):
"What programming languages dominate trending AI repos right now?"
"Is there anything blowing up around MCP or agentic AI this month?"
"Compare the top 5 repos — what do they have in common?"
"Are there any surprise breakouts — repos nobody knows yet but gaining fast?"
"Show me a bar chart of the top 10 repos by stars gained."

That last one triggers Data to Chart >> your agent generates a visualization inline.

What You Built Tonight

"Some of the strongest engineers I've ever met are now handing over almost all their coding to AI."
— Dario Amodei, CEO Anthropic

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- **MCP-Connected Agent**
- Built from scratch
- CoCo as co-pilot
- Yours to keep

107M

- **Real GitHub Events**
- 30 days of data
- No ETL needed
- public S3, free

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- **SQL Written by Hand**
- CoCo wrote every
- query, view, agent,
- and MCP Server

∞

- **Questions**
- From any MCP client
- Claude / Cursor / CoWork
- Same stack, any data

Let's **Stay Connected**

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MEDIUM

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RICHIE BACHALA · SR. MANAGER, SOLUTIONS ARCHITECTURE · SNOWFLAKE

Thank You

Workshop materials + checkpoints:
github.com/sfc-gh-rbachala/building-ai-agents-with-coco-workshop

Free Snowflake trial: https://bit.ly/3SETPFF